INCH-POUND

MIL-C-52648D(ME)
3 December 1993
SUPERSEDING
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MILITARY SPECIFICATION

CRANE, CRAWLER MOUNTED, 40 TON AT, 12 FOOT RADIUS, 2 CUBIC YARD, DIESEL-ENGINE-DRIVEN

This specification is approved for use within the USA Belvoir Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a diesel-engine-driven, crawler-mounted crane suitable for use as a lifting crane with the capability of being converted for clamshell, dragline, shovel, backhoe, wrecking ball, or piledriver operation. The cranes provided shall have a rated lifting capacity of 40 tons at a 12 foot radius.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards. and handbooks form a Part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US ARMY BELVOIR RDE CTR, ATTN SATBE TSE, 10101 GRIDLEY RD STE 104, FT BELVOIR VA 22060-5818 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

A-A-52015	Tagline, Crane, Spring-Loaded, Reel Type.
RR-W-410	Wire Rope and Strand.
TT-V-121	Varnish, Spar, Water Resisting.
VV-F-800	Fuel Oil, Diesel.
PPP-B-601	Boxes, Wood, Cleated Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-P-40	Packaging and Packing of Hand Tools; Tools and Tool Accessoires for Power-Driven; Metal and Wood Working Machinery.

MILITARY

MIL-P-116	Preservation, Methods of.
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173	Varnish, Moisture and Fungus Resistant (For Treatment of Communications, Electronic and Associated Equipment).
MIL-P-514	Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	Treatment and Painting of Materiel. ".
MIL-L-2104	Lubricating Oil, Internal Combustion Engine, Combat/Tactical Service.
MIL-L-2105	Lubricating Oil, Gear, Multipurpose.
MIL-G-3859	Grease Gun, Hand-Operated, Lever, Push and Screw Type.
MIL-G-10924	Grease, Automotive and Artillery.
MIL-L-11025	Lead, Pile-Driver; with Catwalk.
MIL-P-14105	Paint, Heat Resisting (Steel Surfaces).
MIL-B-16028	Bucket, Clamshell.
MIL-B-16029	Bucket, Dragline.
MIL-E-16298	Electric Machinery Having Rotating Parts, Accessories and Associated Support Items, Packaging Of.
MIL-E-17555	Electronic and Electrical Equipment Accessories, and Provisioned Items (Repair Parts); Packaging Of.
MIL-H-21103	Hammers, Piledriver, Self-Powered, Diesel.
MIL-B-22191	Barrier Materials, Transparent, Flexible, Heat Sealable.
MIL-B-23276	Balls, Wrecking.
MIL-H-46170	Hydraulic Fluid, Rust Inhibited, Fire-resistant, Synthetic Hydrocarbon Base.
MIL-T-52176	Teeth, Surface Ripping; (Points, Shanks, Keepers and Keeper-Locks) Bucket and Dipper, Crane Shovel.
MIL-A-52363	Air Cleaner, Intake, Dry Type (for Internal Combustion Engine).
MIL-C-52950	Crates, Wood, Open and Covered.
MIL-C-53039	Coating, Aliphatic Polyurethane, Single Component, Chemical Resistant.
MIL-P-62669	Primer Coating, Synthetic, VOC Compliant (For Brake Drums).

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MIL-V-81940	Valve, Sampling and Bleed, Hydraulic, Type H Systems.
MIL-T-83133	Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment.
MIL-STD-130	Identification Marking of US Military Property.
MIL-STD-193	Paint Procedures and Marking for Vehicles, Construction Equipment and Material Handling Equipment.
MIL-STD-209	Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-810	Environmental Test Methods and Engineering Guidelines.
MIL-STD-838	Lubrication of Military Equipment.
MIL-STD-889	Dissimilar Metals.
MIL-STD-1366	Material 'Transportation System Dimensional and Weight Constraints.
MIL-STD-1410	Methods of Selection of Industrial Engines for End Item Application.
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	Noise Limits for Military Materiel (Metric).
MIL-STD-1791	Designing for Internal Aerial Delivery in Fixed Wing Aircraft.
MS4207	Coupling, Grease Gun, Hydraulic 'Type Nozzle.
MS35644	Filler Neck Assembly, Fuel Tank-Military Vehicles.
MS53052	Bracket Assembly, Liquid Container, Five Gallon.
MS53063	Indicator, Air Cleaner, Intake, Restriction Mechanical Type.

HANDBOOK

MILITARY

MIL-HDBK-113	Guide for the Selection of Lubricants, Fluids, Preservatives and Specialty Products for Use in Ground Equipment Systems.

(Unless otherwise indicated. copies of federal and military specifications, standards, and handbooks are available from the STDZN DCMNT ORDER DESK, BLDG 4D, 700 ROBBINS AVE, PHILADELPHIA PA 19111 -5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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DEPARTMENT OF LABOR (DOL)

29CFR 1910- Occupational Safety and Health Standards (OSHA).
29CFR 1926- Safety and Health Regulations for Construction (OSHA).

DEPARTMENT OF ENERGY (DOE)

10 CFR 40 - Domestic Licensing of Source Material (NRC).

(Application for copies should be addressed to: SUPT OF DCMNTS, GVT PRINTG OFC, WASHINGTON DC 20402.)

US ARMY TANK-AUTOMOTIVE COMMAND (TACOM)

Report No. CR-82-588-003 - STE/ICE Design Guide for Vehicle Diagnostic Connector Assembly.

DRAWINGS

AT

12258941	Connector Receptacle 54 PIN.
12258955	Vehicle Test Card.

(Copies of the TACOM report and drawings required by contractors in connection with specific acquisition functions should be obtained from the USA TANK AUTOMTV CMND, ATTN AMSTA MD, WARREN MI 48397-5000 or the contracting officer.)

ME

13229E8014 - Bracket Assembly Rifle Stowage.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA BELVOIR RDE CTR, ATTN SATBE FMT, 10101 GRIDLEY RD, STE 104, FT BELVOIR VA 22060-5818.)

2.2 Non-Government Publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS, INC. (ASME)

Boiler and Pressure Vessel Code, Section IX Qualification Standard for Welding and Brazing Procedures, Welders, Brazers and Welding and Brazing operators.

B30.5 - Safety Code for Crawler, Locomotive, and Truck Cranes.

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(Application for copies should be addressed to the AMERCN SCTY OF MECHL ENGRS, 345 E 47TH STRET, NEW YORK NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging.
D 3953 - Strapping, Flat Steel and Seals.
D 4675 - Selection and Use of Flat Strapping Materials.

(Application for copies should be addressed to the AMERCN SCTY & MTRLS, 1916 RACE STRET, PHILADELPHIA PA 19103-1187.)

AMERICAN WELDING SOCIETY, INC. (AWS)

AWS D1.1 - Structural Welding Code - Steel.
AWS D14.3 - Earthmoving and Construction Equipment Welding, Section 5.

(Application for copies should be addressed to the AMERCN WELD SCTY, 550 NW LEJEUNE RD, PO BOX 351040, MIAMI FL 33135.)

POWER CRANE AND SHOVEL ASSOCIATION (PCSA)

Standard No. 4 - Crane/Excavator.

(Application for copies should be addressed to the PWR CRANE & SHOVEL ASS, MRNE PLZ STE 1700, 111 E WISCONSIN AVE, MILWAUKEE WI 53202.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC., (SAE)

J154 Operator Enclosures Human Factor Design Considerations.
J159 Load Moment System.
J185 Access Systems for Off-Road Machines.
J209 Instrument Face Design and Location -for Construction and Industrial Equipment.
J220 Crane Boomstop.
J371 Drain, Fill, and Level Plugs for Earthmoving Machinery.
J375 Radius-of-Load or Boom Angle Indicating Systems.
3376 Load Indicating Devices in Lifting Crane Service.
J429 Mechanical and Material Requirements for Externally Threaded Fasteners.
J478 Slotted and Recessed Head Screws.
J514 Hydraulic Tube Fittings.
J534 Lubrication Fittings.
3537 Storage Batteries.
J551 Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices.
J553 Circuit Breakers.
J554 Electric Fuses (Cartridge Type).
J598 Sealed Lighting Units for Construction, Industrial and Forest Machinery.

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J765	Crane Load Stability Test Code.
J820	Crane Hoist Line Speed and Power Test Code.
J833	USA Human Physical Dimensions.
J881	Lifting Crane Sheave and Drum Sizes.
J898	Control Locations for Off-Road Work Machines.
J899	Operator's Seat Dimensions for Off-Road Self-Propelled Work Machines.
J925	Minimum Service Access Dimensions for Off-Road Machines.
J959	Lifting Crane, Hire-Rope Strength Factors.
J983	Crane and Cable Excavator Basic Operating Control Arrangements.
J987	Rope Supported Lattice-Type Boom Crane Structures.
J1024	Fuel-Fired Heaters - Air Heating - for Construction and Industrial Machinery.
J1127	Battery Cable.
J1362	Graphical Symbols for Operator Controls and Displays on Off-Road Self-Propelled Work Machines.

(Application for copies should be addressed to the SCTY OF AUTOMTV ENGRS, INC, 400 CMNWLTH DR, WARMDALE PA 15096-0001.)

(Non-Government standards and other publications are normal 1 y available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. This specification covers a commercial crane consisting essentially of a crawler mounting, full revolving superstructure, and front-end attachments. Each crane shall be complete and ready for operation. Each crane shall be capable of being converted for use in lifting, clamshell, dragline piledriving, wrecking ball, shovel, and backhoe operations. The crane shall be constructed so that conversion to any one of these uses can be effected in the field without the use of special tools or the necessity of Installing additional shafts, drums, clutches or other parts that should for reasons of practicality be included at time of manufacture. The crane shall be capable of traveling over rough ground under adverse or severe conditions. Except for modifications and special equipment required to conform to this specification, the crane shall be the current commercial production model of a standard product. The crane shall meet EPA emissions standards and PCSA No. 4.

3.2 Compatibility. The crane shall be compatible with front-end attachments as specified in 3.17.

3.3 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

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3.4 Material. Materials used shall be free of defects which would adversely affect the performance of components or of the overall assembly. All materials shall be selected by the contractor and shall be of the same quality used for the intended purpose in-commercial practice (see 6.8).

3.4.1 Material deterioration prevention and control. The crane shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of-corrosion and deterioration that may be encountered in any of the applicable operation and storage environments to which the crane may be exposed.

3.4.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic and electrolytic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.4.3 Cadmium plated fasteners. Cadmium plated fasteners shall not be used unless deemed absolutely necessary (see 6.7.and 6.7.1).

3.4.4 Recovered materials. For the purpose of the requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the crane may be newly fabricated from recovered materials to the maximum extent practicable, provided the crane produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the crane.

3.5 Identical items. All cranes furnished shall be physically and mechanically identical in all respects except with prior written approval of the contracting officer.

3.6 Safety.

3.6.1 Safety. All exposed parts which are subjected to high operating temperatures or which are energized electrically and all moving parts that are a hazard to operating or maintenance personnel, shall be enclosed or guarded. Protective devices shall not impair operating functions. Steel walkways and ladders shall be provided, where necessary, for inspection, servicing and maintenance. Walkways shall be provided with safety tread and handrails. All components shall conform to applicable OSHA 29 CFR 1910, general industry, and OSHA 29 CFR 1926, construction equipment requirements. Electrical equipment shall meet NFPA 70 requirements for their intended environments unless otherwise specified. Neither asbestos nor radioactive material shall be used in the crane. Radioactive material is that material in excess of 0.002 microcurie or greater of activity as defined in 10 CFR 40. Power cabinets and diesel engine units shall have provisions allowing for the installation of a fire extinguisher. The provisions shall be readily accessible and visible.

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3.7 Human engineering. The crane, assemblies and subassemblies shall be operable and maintainable by 5th percentile female through 95th percentile male soldiers dressed for the intended environments of use. The crane design shall conform to MIL-STD-1472 design criteria. Design emphasis shall be given to MIL-STD-1472 general requirements, control/display integration, visual displays, audio displays, controls, labeling, anthropometry, environment, design for maintainer, operational ground/shipboard vehicles, and hazards and safety.

3.8 Noise limits. When tested in accordance with 4.5.2.18 and MIL-STD-1474 (5.5), the noise produced by the crane shall conform to MIL-STD-1474 requirements with the exception of 5.1.1.2, 5.2, 5.3, and 5.4. The noise level shall not exceed category C [90 dB(A)]. The provisions of MIL-STD-1474, 4.2 and 4.3 shall be provided if MIL-STD-1474, category D [85 dB(A)] cannot be met. (Warning label, if required, shall specify distance in feet at which 85 dB(A) will be achieved, i.e., “HEARING PROTECTION REQUIRED WITHIN (specify) FEET”).

3.9 Transmrtability. The crane shall be capable of being transported (see 6.2) by rail, and highway transport and be capable of withstanding the shock loads encountered in shipment without damage or permanent deformation. The crane shall meet the transportational, dimensional, and weight constraints plus slinging and tiedown provisions of MIL-STD-1366 and MIL-STD-209. The gross vehicle weight of the crane shall be stenciled on each side of the crane in such a manner as to be readily discernible to handling personnel.

3.9.1 Disassembly and reassembly. When necessary for transport, the crane shall be capable of disassembly and reassembly in not more than 36 manhours using only common hand tools and special tools as furnished with the crane. Cutting and welding on any component or part of the units are not permitted before, during, or after shipment. All major components that must be removed for transport must be equipped with lifting and tiedowns that meet MIL-STD-209 requirements. Small components and accessories removed for transport shall either fit into the tool/storage box or have provisions for stowage on the units from which they were removed.

3.9.2 Tiedown provisions. The tiedown provisions shall conform to MIL-STD-209, class 2 or 3, type III equipment, and to MIL-STD-1791 for equipment restraint criteria. The tiedown provisions shall satisfactorily complete the pull testing as specified in 4.5.2.20 without weld failure, permanent deformation, cracking, loosening, or breaking of the provision or its connecting structural components. Tiedown points shall be labelled “TIEDOWN HERE” and combination lift/tiedown points shall be labelled “LIFT/TIEDOWN HERE”. The labels shall conform to MIL-STD-1472.

3.9.3 Slinging previsions. The slinging provisions shall conform to MIL-STD-209, class 1 or 3, type III equipment. The provisions shall enable the crane to be lifted in the normal operating position. The provisions shall be located so that not less than 1-inch clearance is maintained between slings and all exterior parts and shall be fastened to members which will withstand stresses in the amount and direction of pull specified for the provisions without weld failure, permanent deformation, cracking, loosening, or breaking of the provision

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or its connecting structural components. Slinging provisions may also be used as tiedown provisions when such provisions meet the requirements specified in 3.9.2. Lift points shall be labelled "LIFT HERE". The labels shall conform to the requirements of MIL-STD-1472.

3.9.4 Rail transportability. The crane shall be rail transportable in CONUS and NATO countries without restrictions. When loaded on a 50-inch (127 cm) high rail car, the crane shall have a dimensional profile within the AAR outline diagram and the Gabarit International de Chargement (GIC) or WI, as specified in MIL-STD-1366. The crane shall be capable of withstanding shock loads resulting from rail impact testing specified in 4,5.2.21 without failure, damage, or permanent deformation.

3.9.5 Highway transportability. The crane, when loaded on a semitrailer/ tractor, shall be within the highway permit limits of all states.

3.9.6 Antirotation devices for shipping. Steel rods shall be furnished for securing the revolving superstructure to the track frame to prevent rotation during shipment. One-end of each rod shall be permanently affixed to the track frame. Means shall be furnished to secure the loose end of the rod to the superstructure for use as an antirotation device and to the track frame for storage.

3.10 Maintainability. The crane design, together with component and accessory location and installation, shall permit ready accessibility in accordance with SAE J925 and 3.7, to all items requiring periodic maintenance or service in the field. Covers, safety guards, and plates which must he removed for component adjustment or servicing shall be equipped with quick-disconnect fastenings. Ease of maintenance provisions shall incorporate features assuring clearances for tools and servicing at temperatures specified in 3.11.1. All maintenance, assembly or disassembly operations required shall be accomplished using common tools and special tools furnished with the crane.

3.10.1 Filling, draining, and checking provisions. The transmission housings and other enclosures, such as gearcases, chain cases and reservoirs which contain lubricant or hydraulic oil shall be equipped with dipsticks, sight gauges or check plugs conforming to SAE J371 to determine the fluid levels. Dipsticks shall be etched, knurled, sand blasted, or phosphate coated to facilitate determining the liquid level. Each enclosure shall be equipped with means for filling and draining. The drains shall be fitted with magnetic drain plugs. Each drain plug or valve shall be located so that removal of the plug or valve will result in complete drainage of the fluid from the enclosure (except torque converter). When the crane is in a level position, drainage of fluids shall be into a 2-gallon bucket. Accessibility to the filling means, the fluid-level checking device, and the drain plug or valve shall be provided without removal or adjustment of accessories and parts, except for plates equipped with hand-operable, quick-disconnect fasteners, or capscrews in applications where quick-disconnect fasteners are subject to damage. The drain outlets for the engine-cooling system shall provide complete drainage of the system. The radiator drain shall be such

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that coolant shall not drain onto any part of the crane. Extension hoses are permitted.

3.11 Environmental requirements.

3.11.1 Operational temperature requirements. The crane shall perform as specified herein in any ambient temperature from -25 to +120 °F.

3.11.2 Elevation conditions. The crane shall perform as specified herein at any elevation from sea level (barometric pressure 29.92 inches mercury) and a maximum ambient temperature of 120 'F to 5,000 feet above sea level (barometric pressure 24.89 inches mercury) and a temperature of 107 'F.

3.12 Performance.

3.12.1 Capacities.

3.12.1.1 Crane capacity. The crane, when used as a lifting crane, shall have rated lifting capacities through 360 rotation not less than those specified in table I.

TABLE I. Rated crane lifting capacities.

Boom length (ft)	Radius of load (ft)	Rated load (lb)
40	12	80,000
60	50	10,000

3.12.1.2 Shovel capacity. The crane, when equipped with a shovel attachment (see 6.2), shall have a shovel dipper capacity not less than that specified in table II. The working heights, radii, and digging depth of the shovel shall be not less than those specified in table II with the crane shovel resting on firm level ground and the shovel boom positioned at a 45 degree angle.

TABLE II. Shovel operating range.

Shovel dipper capacity	Dumping		Cutting		Digging depth below floor level	Clean radius floo leve
	Height door open	Radius max. height	Height	Radius		
(cu yd)	(ft) (in)	(ft) (in)	(ft) (in)	(ft) (in)	(ft) (in)	(ft) (
2	17 6	28 0	17 0	32 0	7 0	21

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3.12.1.3 Backhoe capacity. The crane, when equipped with a backhoe attachment (see 3.18.2.11 and 6.2), shall have a capacity, bucket width, ground clearance at start of dump and the digging depth of not letters than those specified in table III with the crane shovel resting on firm level ground.

TABLE III. Backhoe working ranges.

Capacity	Ground clearance at start of dump		Digging depth		Bucket width w/o side cutters (min)
(cu yd)	(ft)	(in)	(ft)	(in)	(in)
2	12	0	25	0	50

3.12.1.4 Clamshell and dragline capacity. The crane shovel equipped with the basic boom shall be adaptable to and operable with a 2-yard capacity clamshell bucket and with a 2-yard capacity dragline bucket.

3.12.1.5 Crane capacity plate. The crane shovel shall be provided with a crane-capacity plate mounted inside the cab in a location conspicuous to the operator. The plate shall contain the following information: .

a. Rated loads in 5-foot increments for the full range of operating radii, angles and superstructure positions when using the basic boom length specified in table I and for each 10-foot increment in boom lengths up to and including 160 feet. This plate shall also include capacity information for jib lengths as furnished.

b. Indication that the rated loads are loads at a radius measured from the center of rotation of the revolving superstructure to the centerline of the hook block and that the weights of all lifting means except the hoist rope are to be included as part of the load.

c. Statement that the rated loads are based on the crane standing on a supporting surface that is firm, level, and uniform.

d. Statement of deduction to be made from the rated loads for components of crane attachments such as jibs.

3.12.2 Operating characteristics. All operational functions for a given front-end attachment shall operate independently and simultaneously for maximum performance.

3.12.2.1 Maneuverability. The crane shall be capable of traveling in a forward or backward direction with the revolving superstructure in any position, and capable of being steered either to the right or left with controls from the operator's position. The completely equipped crane with applicable front-end attachment located either to the front or back of the crane shall conform to the following mobility requirements:

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a. Capability of ascending, descending, and stopping on a 30 percent grade not less than 35 feet in length on smooth, firm, dry ground.
b. Capability of negotiating a 15 percent side slope on smooth, firm, dry ground.
c. Capability of travel on firm, level, dry ground with a maximum speed of not less than 0.75 miles per hour.
d. Ground clearance not less than 12 inches.
e. Capability of fording and operation on hard-bottomed bodies of water up to 12 inches without evidence of damage or water leakage.

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3.12.2.2 Ground bearing pressure. The ground bearing pressure of the crane with basic boom, as calculated in 4.5.2.4, shall be not greater than 15 pounds per square inch (psi).

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3.13 Backward stability. The center of gravity from the axis of rotation shall be not farther than 70 percent of the radial distance from the axis of rotation to the tipping fulcrum in the least stable direction when the crane is resting on firm, level, and uneven supporting surface; in working order without load; and with the basic boom at the minimum working radius.

3.14 Crane load stability. Crane stability under loads shall be in accordance with SAE J765. The load rating chart load capacity shall be not more than 75 percent of the stability established.

3.15 Stress requirements. Design of the completely assembled crane shall be such that the maximum stress imposed shall not exceed the strength margins contained in SAE J987. Tests in accordance with SAE J987 shall have been performed on the model crane being provided, and test results shall be made available for Government inspection at the time of first article tests.

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3.16 Superstructure. The superstructure shall consist essentially of a weather tight cab enclosing the operating machinery and operator's position, mounted on a full revolving platform. The principal members of the platform shall be designed to receive the front-end attachments with mounting pins. Means shall be provided for retaining the pins. The pins shall be capable of being aligned with common hand tools. The machinery deck shall be sealed with boots or grommets where controls pass through the deck. Two clips shall be furnished and fastened on the front edge of the operating frame to provide connection points for the inner end of the piledriver lead catwalk (see figure 1). Access systems shall conform to SAE J185. Minimum access dimensions shall conform to SAE J925. A ladder shall be attached to the superstructure cab to facilitate accessibility to the gantry and A-frame sheaves. The fuel tank filler neck, made in accordance with MS35644, shall be accessible and no more than 36 inches from the ground or from an antiskid platform space not less than 15 inches square, to enable the operator to fuel the crane tank with 5-gallon cans or to aid in bulk fueling. A catwalk conforming to SAE J185 shall be installed around the superstructure except for the boom foot area. The catwalk shall be bolted to the superstructure and shall be removable. Access footholds or handholds to enable the operator to gain access to all platforms and catwalks shall be provided. Removable counterweights shall be provided. The complete counterweight shall be capable of removal and placement on

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a trailer or truck in not more than 2 hours by two men using only common hand tools and the lifting provisions of the crane. The complete counterweight shall be capable of replacement on the crane by two men in not more than 2 hours using the same tools and the lifting provisions of the crane. The counterweight may serve as the rear catwalk providing the walking surface is an antiskid surface and otherwise meets all requirements specified herein.

3.16.1 Functional requirements. With the engine operating at governed speed, the crane superstructure shall meet the following requirements:

a. When tested in accordance with SAE J820, hoisting-drum single line power of not less than 3,500,000 foot-pounds (ft-lb) per minute with bare drum (single layer).
b. A swing speed of not less than 3.5 revolutions per minute (rpm) at governed engine speed.
c. Hoisting-drum single line speed between 147 and 180 feet per minute (fpm) at governed engine speed.

3.16.2 Forward machinery deck protection. A sheet metal trough shall be provided to protect the forward machinery deck from mud and dirt scraped from the haul-in cable during dragline operations. The trough shall not interfere with any of the specified operations.

3.16.3 Cab. The cab for the crane shall be fabricated from sheet steel or fiberglass, having sufficient windows and glazed doors to permit at least 270 degrees lateral visual field for both the crane and carrier operator and 15 degrees above horizon visual field for the operator of the carrier. All glass shall be safety type, and shall be held in place by rubber or metal channels for easy replacement. Windows shall be arranged for easy opening and removal when required, All cab doors, whether of the sliding or swinging type, shall be adequately restrained from accidentally opening or closing when the crane is traveling or operating. All doors and opening windows shall be provided with means for securely locking in the closed position. Adjustable and comfortable seats shall be provided in each cab, located to provide maximum unobstructed visibility for the operator. The cab shall be equipped with air conditioning, heater, defroster, and air or electric operated windshield wiper. The wiper shall have at least a two speed control and a positive lock park position. An electric or air horn, hand or foot operated, shall be provided. A heater with two or more fan speeds and a defroster shall be furnished. The heater shall be capable of reaching and maintaining a uniform cab temperature of 41 'F ($\pm$ 9 °F) or higher at an ambient temperature of -25 °F within 1 hour. The defroster may be either part of a heater-defroster or a separate defroster fan and shall provide defrosting of the front and side windows. If a fan type defroster is provided, it shall be safety guarded, adjustable, and provided with an on/off switch. If the heater is fuel fired, it shall conform to SAE J1024. The cab shall be provided with adequate lighting/illumination to enable the operator to read all gauges, charts and control labels under conditions of darkness. The cab shall have provisions for storing the operator's M-16 rifle in a rack. The cab shall completely enclose the operator's seat, control levers, and pedals, instrument panel, and all main-deck machinery in accordance with SAE J154. Hinged covers with slots for

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hoist cables shall be provided for weather protection of the cable drums. The seat, instruments, and controls shall be in accordance with SAE J833 and SAE J898. The work area, including the boom point sheaves at the minimum operating radius, shall be visible to an operator varying from large to small in accordance with SAE J833. All doors shall be held open and closed with a positive locking device and shall be provided with latches to prevent accidental opening and closing. The seat, conforming to SAE J899 shall be located and adjustable so that an operator ranging in sizes as indicated in SAE J833 is within reach of the control levers and pedals. Walking surfaces and steps shall be of the antiskid type. Drainage provisions shall be made for any wells or depressions that are subject to collection of water.

3.16.3.1 Instrument panel. An instrument panel containing all switches, gauges, and instruments, except the hour meter (which may be mounted on the engine for complete control of the engine and lights), shall be located in accordance with SAE J209, and be placed within reach and visibility of the operator in the normal operating position. The instrument cluster group shall be removable to permit access to all wiring and units. Panel lights shall be furnished for visual reading of all instrument, indicators, and identification plates by the operator. Each switch, control, or gauge on the instrument panel shall be labeled in accordance with SAE J1362. Arrows or any appropriate mark showing the direction to operate, if pertinent, shall be provided. Identification plates furnished shall conform to 3.20. The tachometer shall be accurate with plus or minus 5 percent of actual engine speed. The panel shall contain the manufacturer's standard array of controls and instrumentation but shall include no less than the following controls and instruments:

a. Engine start/stop speed control.
b. Engine revolutions per minute indicator (rpm).
c. Engine operating temperature gauge.
d. Oil pressure gauge.
e. Battery charging ammeter/voltmeter.
f. Emergency cut-off control.
g. High temperature cut-off switch.
h. Fuel gauge.
i. Low oil pressure cut-off switch.
j. Oil filter.
k. Air cleaner, dry type.
l. Engine hour meter.
m. Unassisted cold start control.

3.16.3.2 Superstructure cab, disassembly. For transportability, the superstructure cab shall be capable of removal to reduce the size, weight, and shape of the cross section of the crane. The framing shall tie the side panels and main portion of the roof together as one unit. The top corners shall be provided with lifting eyes to accommodate crane slings. The supporting members of the cab shall be bolted to the machinery deck with fasteners of sufficient strength and quantity so as to hold the cab securely in place under all operating conditions. All free-standing edges of the removable and fixed section shall be provided with stiffeners and self-sealing, labyrinth-type weather seals to keep

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out water. All electrical, air, and hydraulic lines fastened to the walls of the cab shall be equipped with quick-disconnect, self-sealing couplings as needed at the separation line of the cab and the machinery deck.

3.16.4 A-frame. The A-frame or gantry provided shall be capable of withstanding stresses induced by lifting a 160-foot boom from the ground to operating position. The principal vertical compression and tension members of the A-frame shall be connected so that the operating stresses are transferred to the main structural members of the rotating frame of the superstructure. The gantry shall be of the folding or retractable type for overhead clearance or reducing to cube during transportation.

3.16.5 Power load lowering. For crane operations, the primary load hoisting drum shall be equipped with a powered reversing drive and suitable control features to provide precision controlled power lowering of loads by the main hook. The powered load lowering mechanism shall be controlled by a lever conveniently located within the revolving superstructure, and shall be capable of lowering all rated loads with the manufacturer's recommended reeving. The powered load lowering mechanism shall be capable of lowering the loads at a smooth, steady controlled rate of speed; and a smooth variable controlled rate of speed; dependent on engine speed, without jerking, bouncing, or slipping the load. A braking device may be used to assist the power lowering mechanism. Extended use of the powered load lowering mechanism shall not result in clutch or brake failure by fading, grabbing, cracking, internal rupture, breakage, permanent deformation, or any malfunction. Clutch and brake lining temperatures shall not exceed the maximum recommended by the manufacturer of the lining. The powered load lowering mechanism shall be capable of being disengaged, when not in use, so as not to interfere with normal crane operations.

3.16.6 Crane-boom hoist. The crane-boom hoist shall be independently controlled and shall safely suspend the boom with recommended reeving without attention from the operator, shall provide for controlled lowering under the operator's control, and shall make free spooling of the boom cable impossible under any circumstance. The boom hoist shall raise and lower the boom with rated capacity loads. It shall have sufficient rope capacity to allow the boom tip to be lowered to ground level with the crane resting on level ground and not less than three wraps of wire rope retained on the drum. The boom hoist shall be equipped with a positive lock to facilitate holding the boom in any set position. The boom-hoist drum shall be arranged to minimize eccentric pull in order to avoid chafing of the rope and unequal wear on the drum shaft bearings. The drum shall spool the rope in even layers. The boom hoist clutch shall be readily accessible for adjustment and repair.

3.16.7 Drums and sheaves. The drums shall be independently driven and suitable for use with any of the different front-end attachments and arranged so as to reduce eccentric pull to a minimum in order to avoid chafing of the rope and unequal wear on the drum shaft bearings. The complete drum assembly and mounting shall rotate freely when clutches and brakes are not engaged to allow gravity lowering of the hook block. The drum shall be equipped with a socket and wedge for dead-ending the wire rope. Lubricating fittings shall be located to permit

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Lubrication without unspooling cable. Drums shall have sufficient rope capacity to accommodate the required rope lengths. Pitch diameters of drum and drum lagging including the boom hoist drum shall be in accordance with SAE J881. The machinery operating the primary and secondary drums shall be arranged so that with the engine clutch disengaged, the drum clutches may be simultaneously engaged, to reverse the operating machinery by lowering the load on either drum under control of both drum brakes simultaneously for a precision load lowering without the boom being raised. All drums shall be guarded to prevent the rope from leaving the drum during operation. The pitch diameter of load bearing sheaves shall be in accordance with SAE J881. Flanges on the boom point sheaves shall extend beyond the rope grooves not less than 1.75 times the diameter of the rope. All sheaves shall be guarded to prevent the rope from leaving the sheave groove during operation. The sheave pins shall be of sufficient size and strength to support the sheaves under maximum operating conditions.

3.16.7.1 Drum bakes. The drum brakes shall hold and slowly lower (without jerking or chattering), capacity loads supported by recommended reeving. Provision shall be made for locking the brakes in any position of the drum. The drum brake shoes shall be easily removable for lining replacement.

3.16.7.2 Drum clutches. Drum clutches shall be capable of withstanding rated speed and continuous operation without fading, accessible for adjustment and repair, and sufficiently large to eliminate slippage under 133 percent of rated load with recommended reeving. The primary and secondary load hoist drums shall be operated by clutches controlled by levers and shall be designed so that once the clutch is engaged, it shall be held in place to hold the clutches in engagement and be able to perform the power lowering tests specified in 4.4.9. The clutches shall be unlocked by pulling or pushing forward on the control lever.

3.16.7.3 Auxiliary drum. When specified (see 6.2), an auxiliary or third drum shall be provided and shall meet the requirements of 3.16.7, 3.16.7.1, and 3.16.7.2.' The drum shall be engaged by a clutch operable from a control lever in a position within easy reach of the operator while seated.

3.16.8 Rollers. The rotating frame shall be supported by fixed rollers, or by a live ball or roller ring. Rollers, balls, roller rings, and ball races shall be heat-treated hardened steel and shall be accurately machined. Fixed rollers shall be fitted with anti-friction bearings and shall be provided with grease fittings. The balls or rollers of live ball or roller rings shall be accessible and shall be removable. When the hold-down function is performed by hook rollers, such rollers shall be either the dual-purpose type in which each set of rollers performs both the supporting and hold-down functions, or the rollers may be of the hold-down type only. Hook rollers shall operate either singly or in pairs on tandem bogeys and shall be provided with grease fittings. Hook rollers shall be provided with means for vertical or horizontal adjustment as required. The hook rollers shall be removable without removal of the hook-roller brackets from the revolving superstructure. The hook roller brackets shall not interfere with the removal of the superstructure from roller path and frame. When tested in accordance with 4.5.2.7, the temperature of the supporting balls or rollers and hook rollers shall not exceed 175 'F.

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3.16.9 Swing mechanism. The rotation of the rotating frame shall be controlled through clutches designed for fast cycling, craning, and other operations, and capable of transmitting not less than 75 percent of the maximum engine torque. The temperature of the swing clutches shall not exceed the maximum temperature recommended by the clutch lining manufacturer. The swing speed at governed engine speed shall be not less than 3.5 rpm.

.3.16.9.1 Swing-gear, lock, and brake mechanisms. The revolving superstructure swing shall be actuated by a gear and pinion. A swing lock shall be provided to lock the revolving superstructure in any desired position. A swing brake shall also be provided to control the swing of the revolving superstructure and to hold the superstructure from swinging when the crane is working on a slope and shall require manual effort to engage. Both lock and swing-brake controls shall be located within reach of the operator while seated.

3.16.10 Power transmission. The engine shall be connected to the transmission through a friction disengaging clutch and a hydraulic torque converter. The clutch and torque converter shall be located so as to be accessible for adjustment and repair.

3.16.10.1 Clutch. The clutch shall be controlled from the operator's position and shall be capable of being engaged and disengaged instantly while the engine is running.

3.16.10.2 Drive chains. All drive chains shall be provided with a means of adjustment. This adjustment shall be made either by changing the center to center distance between the driving and driven shafts or by means of a roller or sprocket meshing with or riding on the chain. This roller or sprocket shall be located so that tension shall be exerted on the exterior side of the slack side of the chain. The primary drive chain between the torque converter output shaft and the first sprocket shall be encased. The case for this chain shall provide a leak-proof lubricant reservoir.

3.16.11 Electrical system. A standard 24-volt negative ground electrical system with a 24-volt starting motor and a heavy duty battery charging alternator shall be furnished.

3.16.11.1 Electromagnetic Interference (EMI) suppression. Electromagnetic radiation from the crane shall be within the limits of SAE J551. When required, the crane shall be equipped with radio noise suppressors conforming to SAE J551. The horn is exempt from the EMI requirement.

3.16.11.2 Wiring. The wiring of the crane shall be fabricated and installed to provide a reliable electrical system. Conduit shall be furnished for all circuits to within 12 inches of components, with the exception of the battery cables. Conduit, clips, loom, grommets, and cushioned support clamps shall prevent whipping, chafing, or other mechanical damage to the wiring and harness. Wiring shall be routed within the framing members whenever possible. Removal of major assemblies such as engine, cab, and instrument panel cluster shall be accomplished by disconnecting one or more plugs and receptacle with the exception of battery,

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starter, and slave cables. All circuits shall be protected by circuit breakers in accordance with SAE J553 or fuses in accordance with SAE J554.

3.16.11.3 Lights. Lights shall be controlled by switches, and be protected by circuit breakers. Switches shall be located on the instrument panel in the revolving superstructure within reach of the operator when seated. One switch and associated electrical circuit shall be provided for the independent control of the following light combinations, as furnished:

a. Revolving superstructure floodlights.
b. Interior cab lights.
c. Instrument lights.

3.16.11.3.1 Floodlights. Not less than six floodlights, conforming to SAE J598 shall be furnished and shall be attached to the crane on mounting brackets providing vertical and horizontal adjustments. The housings shall be mounted in a position to permit replacement of the lamp. Lamp housings shall not extend beyond the side of the revolving superstructure unless provision is made for folding the lamps clear of the sides of the superstructure for transport. The floodlights shall be located so that lenses shall not shine directly into the operator's eyes when the operator is working in the normal operating position. The floodlights shall adequately illuminate the working area so that typical work may be performed at night. Not less than four floodlights, two on each side of the boom hoist cables, shall be mounted at the top front edge of the cab to illuminate the working area. Not less than two floodlights, one on each side of the boom mounting shall be mounted as near the side and bottom of the superstructure as practicable.

3.16.11.3.2 Interior lights. The cab shall have not less than one interior ceiling light with plastic lens not less than 4 inches in diameter. The light(s) shall be effectively located. Each light shall be equipped with a bayonet type bulb. The overall depth of the light fixture from the face of the lens to the ceiling shall not exceed 3 inches.

3.16.11.3.3 Trouble lamp. A handgrip type, portable, trouble light shall be mounted in the cab adjacent to the operator's door. The trouble light shall have a handle incorporating an on-off switch. The trouble-light mounting bracket shall protect the sealed beam unit from damage.

3.16.11.4 Warning horn. The revolving superstructure shall be equipped with a warning horn. The horn shall be mounted in a protected location on the exterior of the revolving superstructure. The horn shall be operated by a push-button switch mounted within reach of the operator.

,3.16.11.5 Starting systems. The engine starting systems shall be as recommended by the engine and starter manufacturers for this application. The starting systems shall include a starter switch, engine speed disconnect, main starter solenoid, starter motor and flywheel drive mechanism.

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3.16.11.6 Charging system. An engine driven battery charging system shall be furnished. Adjustable generator or alternator mounting brackets shall provide movement for adjusting belt tension, and shall resist belt tension forces to minimize potential loss of recommended belt tension.

3.16.11.7 Fungus and moisture resistance. The electrical circuitry, including all components and connections shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173.

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

The varnish shall be applied to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.16.11.8 Batteries. Batteries shall conform to SAE J537 and shall be readily accessible for servicing.

3.16.11.9 Battery enclosure. A weather protected compartment type enclosure with a battery quick release shall be provided and be accessible for servicing and maintenance without removal of the battery(s). The enclosure shall be equipped with electrolyte resistant battery holddown components to prevent movement or damage to the batteries. The enclosure shall include a drain for water or electrolyte, and a vent to prevent gas accumulation. Battery cables conforming to SAE J1127 shall be furnished with insulated terminal covers. Positive and negative cable terminals shall be identified with a red sleeve labeled "+" and a black sleeve labeled "-", respectively, and corrosion resistant bolts and nuts provided. Negative ground shall be provided. The battery enclosure shall be of sufficient size to accommodate the military type 6TL or commercial type 8D battery.

3.16.12 Diesel engine. The diesel engine, including all systems and accessories, shall conform to MIL-STD-1410, class I, including EPA emission standards and as specified herein. Each diesel engine shall be fully enclosed and provided with access doors to permit operation, service, and maintenance as required. The diesel engine shall meet the performance requirements specified herein, including unassisted cold start operation, when operating with fuel oil conforming to VV-F-800 and turbine fuel conforming to MIL-T-83133, grade JP-8. Fuel tanks shall not allow spills or overflows to run onto the engine, exhaust, or electrical equipment. The tank shall be mounted in a protected location near the outside frame member and be accessible for refueling without obstruction. When the fuel tank is located at a level which is higher than the engine fuel intake, a fuel shutoff valve shall be installed in the fuel supply line to permit engine service or maintenance without draining the fuel tank. Fuel tanks shall be corrosion protected and conform to the special finish requirements of MIL-STD-193

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and shall have military standard filler necks in accordance with MS35644. Fuel lines shall be adequately supported and shall be protected against mechanical damage. Grommets shall be used where fuel lines pass through frames, bulkheads, or areas where abrasive action could occur. Construction and mounting shall be such that neither vibration nor tightening of mounting devices shall cause leaks to develop. The filler cap shall not protrude from the body of the system and fuel type shall be stenciled near fuel cap in letters not less than l-inch letter height. A fuel tank drain plug located at the lowest point shall be furnished. The capacity of the fuel tank shall be sufficient for 8 hours of continuous operation at 75 percent of the rated intermittent brake horsepower.

3.16.12.1 Oil sampling valve. An oil sampling valve conforming to MIL-V-81940 shall be provided on the engine. The oil sampling valves shall be labeled adjacent to the valve indicating the type of oil sampled ("ENGINE OIL SAMPLE"). The valves shall be manually operated and shall automatically close after release. The valves shall be made of material resistant to corrosion that shall not contaminate the sample. The discharge port of the valves shall be covered with a captive chained cap conforming to SAE J514. The valves shall be located to ensure that personnel shall not be exposed to danger when taking oil samples with the engine running. The location of the oil taps shall be such that when samples are taken, it shall be a true representation of oil that is flowing while the engine is running.

3.16.12.2 Diagnostic Connector Assembly (DCA). An easily accessible Diagnostic Connector Assembly (DCA) for interface with the Simplified Test Equipment for Internal Combustion-Engines Reprogrammable (STE/ICE) test equipment shall be furnished. The DCA shall be in accordance with TACOM Drawing 12258941 and shall be labeled "Diagnostic Connector Assembly". All requirements for DCA shall be in accordance with STE/ICE-R design guide for vehicle diagnostic connector assemblies, report number CR-82-588-003, revision L The DCA shall have the capabilities for measuring the minimum functions defined in table 2-2 of the design guide. Additional test parameters to be monitored in either the DCA or transducer kit (TK) mode shall be as follows: fuel return pressure, engine oil pressure, engine oil filter pressure drop, starter solenoid voltage, and transmission oil pressure. Test points identified to monitor functions in the TK mode shall be accessible without the removal of major assemblies and installed attachments. Those additional test parameters which are inaccessible for monitoring without the removal of other components in the TK mode shall be monitored in the DCA mode. Fittings/adapters required for interface with the STE/ICE-R test set (when monitoring in the TK mode), shall be permanently installed to ensure that adapters in the STE/ICE-R test set readily mate with the TK mode test point locations. A separate wiring harness shall be provided for the DCA and shall include all wiring and necessary hardware to perform required capabilities. A fuel shut-off method shall be provided for conducting compression unbalance checks. The contractor shall also provide STE/ICE-R vehicle test cards in the format specified in TACOM Drawing 12258955 addressing both DCA and TK measurements, and shall incorporate the test cards into the technical manuals.

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3.16.12.3 Operation conditions. All systems, including fuel system, lubrication system, air induction system, and coolant system, shall perform without leakage or malfunction when the crane is negotiating a 30 percent grade or a 15 percent side slope.

3.16.12.4 Air cleaner. The air cleaner shall be of the dry type conforming to MIL-A-52363, class III. The air cleaner shall be provided with a restriction indicator conforming to MS53063. The air cleaner shall be accessible for maintenance and servicing.

3.16.12.5 Engine exhaust system. An engine exhaust system shall be furnished, and shall expel gases away from the operator's cab.

3.16.12.6 Governor. The engine shall be controlled by a variable speed governor, Selected governor settings through the operating range shall be provided by a control lever which shall be accessible to the operator in the seated operating position.

3.16.13 Operator controls. The crane shall be controlled by the minimum quantity of hand levers and foot pedals necessary for efficient operation. All levers and pedals shall be conveniently located in accordance with SAE J898 and arranged in accordance with SAE J983, f-or easy access and operation by the operator (when seated in the normal operating position) without causing obstruction to the operator's view of the work. All brakes shall be capable of being easily locked and unlocked. Control levers shall be free of all latches which may seriously delay emergency operation. All levers and pedals shall be permanently marked as to the function it controls. When the same lever controls more than one function, all such functions shall be marked. These markings shall be made on plates. The plates shall be located adjacent to the control and visible to the operator.

3.17 Crawler mounting.

3.17.1 Crawler base. The crawler base shall be steel casting or shall be rigidly constructed of heavy steel shapes, fitted and welded to prevent permanent distortion when traveling across sharp depressions and uneven ground. The swing gear and turntable roller path shall be steel and shall be rigidly attached to or integral with the base. The turntable roller or shear ball path shall be of sufficient size and strength to support the revolving superstructure under all conditions without distortion. Where hook rollers perform the holddown function, the lower member of the center pivot bearing shall be integral with, shrunk into, or welded to the crawler base. The center pin or gudgeon shall operate in bronze or anti-friction bearings.

3.17.2 Track assembly. The crawler tracks shall be constructed for operation in loose dirt, mud, rocks, and snow; shall be provided with suitable adjustments for maintaining correct tension in the track; and shall be constructed so that the track cannot be accidentally disengaged. The track shoes shall be wear resistent steel of flat, self-cleaning construction. The track connecting pins shall be steel, of sufficient size, hardness, and strength, for the purpose intended and,

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shall be removable and replaceable. The tracks shall be supported on steel rollers to distribute the weight of the crane over the ground for the entire length of the crawlers without depending upon the tension in the track for support. All track assembly shafts and rollers shall be provided with guarded grease fittings. Where practicable, grease fittings shall be accessible from outside the crawlers and lower base.

3.17.3 Side frames. The crawler track side frames shall be steel castings or welded structural steel members.

3.17.4 Crawler drive. The crawler shall be driven by gears, roller chains, or hydraulic motors. The travel mechanism shall have positive steering controlled from the operator's cab. The crawler clutches shall be of adequate size to permit constant operation without overheating. Brakes or other locking means shall be provided to hold the crane stationary during the working cycle while the crane is on level grade, and to prevent the crane from accidentally rolling down a 30 percent grade. Brakes, that can be applied at any time, shall be furnished on the travel mechanism. It shall be possible to apply the brakes on both tracks at the same time. Jaw clutches, when furnished, shall have a sufficient number of teeth to engage and to transmit maximum traction power without undue wear. The travel gears and bearings shall be protected against the entrance of dirt or other foreign matter. The propelling mechanism shall be provided with a positive traction lock or brake for locking the crawlers. The propelling shaft shall be so arranged that it is not one or a part of the supporting axles. All shafts shall be bushed or provided with anti-friction bearings and ball-check grease fittings.

3.17.5 Axles. Each axle or crosspiece shall be either of cast, forged, fabricated or structural steel.

3.18 Front-end attachments. The revolving superstructure of each crane shall be equipped with all necessary equipment to permit operation as a crane (boom with or without a jib), clamshell, dragline, piledriver, shovel, backhoe, and crane with a wrecking ball. Tools, equipment, parts, and material which are not components of the crane and are required for operation of front-end attachments, shall be furnished with each attachment.

3.18.1 Primary front-end attachments. The following attachments shall be furnished with each crane and such attachments shall be complete with all necessary parts, including wire rope.

3.18.1.1 Crane boom. The crane boom shall be suitable for crane, dragline, clamshell, piledriver, and wrecking ball operation. The basic boom shall be 40 feet in length, and shall consist of two sections. The boom shall be capable of being extended to a maximum length of 160 feet by the use of boom extensions (see 3.18.2.1). The boom point shall be equipped with not less than three guarded parallel sheaves conforming to 3.16.7 and with not less than two independent dead-ends commensurate in size with the wire rope furnished. Piledriver adapter plates conforming to figure 2 shall be provided to adapt the leads specified in 3.18.2.8 to the crane boom. Adapter plate support bracket(s) shall be furnished on the inner section of the boom.

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3.18.1.2 Pendant equalizer assembly. A pendant equalizer assembly shall be furnished.

3.18.1.3 Pendant cable. A matched set of pendant cables shall be provided for connecting the pendant equalizer assembly to the boom point.

3.18.1.4 Boom safety stop. A telescopic boom stop shall be furnished to provide a safeguard against overtopping of the boom. The complete boom stop with supporting attachments shall have sufficient strength to develop the ultimate strength of the boom in bending at the point of attachment of the boom stop to the boom. The boom stop shall provide a shock-absorbing action and shall remain assembled when the boom point is lowered to the ground. The boom safety stop shall conform to SAE J220. Under all operating conditions specified herein, as the boom is raised from the ground to the maximum allowable angle, the boom stop shall telescope freely without sticking, binding or bending.

3.1$.1.5 Boom hoist clutch throwout. Means shall be provided to disengage the clutch of the boom hoist drum when the boom has reached the maximum angle or height and is in contact with the boom stops.

3.18.1.6 Boom angle indicator. An indicator which will show the position of the boom at all times shall be provided. The indicator shall be of the heavy-pendulum type swinging on a double-sealed anti-friction bearing. The pointer shall indicate the angle of the boom with the horizontal plane. The indicator shall be located on the boom, near and visible to the operator's normal operating position. A chart located in the cab near the operator shall be provided showing rated lifting capacities at various boom angles for various boom lengths, with a statement of deductions to be made when using the jib. The boom angle indicator shall conform to SAE J375.

3.18.1.7 Overload indicating system. When specified (see 6.2) a load indicating warning system of the load moment type shall be furnished in accordance with SAE J159, SAE J375, and SAE J376. This system shall automatically indicate and warn the operator when an overload condition exists that exceeds the rated loads specified in 3.12.1.1 through the use of lights, displays, and an audible signal mounted within the operator's view within the cab. The audible and light signals shall cease and displays shall indicate a safe loading condition when corrective action has been taken by the operator. The indicating system may utilize the electrical system of the crane. The electrical system shall be waterproof and shall perform under all operational conditions specified herein without any malfunction.

3.18.1.8 Crane hook block. A commercial crane hook block with safety latch suitable for the application shall be provided.

3.18.1.9 Wire rope. All wire rope except for the tagline shall be IWRC improved plow steel, preformed right regular lay wire rope conforming to RR-W-410. Wire rope strength factors shall be in accordance with SAE J959.

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3.18.1.9.1 Rope sockets. All wire ropes, except tag lines, pendant cables and dipper trip lines, shall have ends attached to drum laggings or to dead-end connections by wedge-type sockets. Clamps, clips, and screw wedge connectors shall be permitted only on tag line and dipper trip lines. Boom pendant cables specified in 3.1$.1.3 shall have swaged open-end sockets, and pendant cables for boom extensions specified in 3.18.2.1 shall have one open-end socket and one closed-end socket.

3.18.1.9.2 Rope lengths. Wire rope shall be of the greatest length required to perform as a crane Performing the functions specified in table IV and based on the maximum boom length furnished. The boom hoist shall have sufficient wire rope to allow the tip of the boom to be lowered to rest on level ground with three wraps of wire rope remaining on the boom hoist drum.

TABLE IV. Crane function to determine rope lengths.

Function	Position of boom	Position of attachment
Crane hook	At minimum radius.	At ground level.
Clamshell	60 degrees from horizontal.	20 percent of boom length (in feet) below ground level.
Dragline	40 degrees from horizontal.	50 percent of boom length (in feet) below ground level and 30 percent of boom length (in feet) beyond boom tip.
Piledriving	At basic boom height.	Two basic boom lengths +50 feet below ground level.
Auxiliary drum hoisting	At maximum boom height.	Two boom lengths +20 feet.

3.18.2 Secondary front-end attachments.

3.18.2.1 Boom extensions. Boom extensions shall be furnished to provide the specified boom length (see 6.2). Extensions shall provide for an increase in the boom length in 10-foot increments and shall be furnished with matched sets of pendant cables. Extensions shall match and mate with the other boom sections.

3.18.2.2 Boom jib. When specified (see 6.2), a boom jib shall be furnished complete, and shall be readily attached to the crane boom (without piledriver leads in place). Boom jib lengths shall be as specified (see 6.2). Boom jib shall be provided with necessary riser and holdback ropes, and complete with adequate overhaul ball and hook.

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3.18.2.3 Clamshell bucket. When specified (see 6.2), a clamshell bucket shall be furnished and shall conform to MIL-B-16028, type II. Bucket teeth shall be in accordance with MIL-T-52176. Side and corner teeth shall be furnished.

3.18.2.4 Tagine. When specified (see 6.2), a tagline shall be furnished and shall be of the reel type, with an adjustable tension mechanism, shall be in accordance with A-A-52015, and shall control the bucket specified in 3.18.2.3. The tagline shall be provided with means of attaching to the boom without cutting, welding, or drilling of the boom and shall be removable and replaceable with common tools. All necessary parts for installations shall be furnished.

3.18.2.5 Dragline bucket. When specified (see 6.2), a dragline bucket shall be furnished and shall conform to MIL-B-16029, type II. Bucket teeth shall be in accordance with MIL-T-52176.

3.18.2.6 Dragline fairlead. When specified (see 6.2), a dragline fairlead shall be furnished, shall be of the four-sheave swinging or two-sheave revolving type and mounted in accordance with the manufacturer's standard procedure. The fairlead shall have either replaceable wearing plates or replaceable guide rollers, shall have bronze-bushed or anti-friction bearing-mounted sheaves with smooth cable grooves, and shall be arranged to avoid chafing the drag cable in any working position.

3.18.2.7 Wrecking ball. When specified (see 6.2), a wrecking ball shall be furnished and shall conform to MIL-B-23276, size as specified.

3.18.2.8 Piledriver leads and catwalk. When specified (see 6.2), piledriver leads and catwalk shall be furnished and shall conform to MIL-L-11025. Lead sections shall consist of one top section and lower sections in numbers as required (see 6.2).

'3.18.2.9 Piledriver hammer. When specified (see 6.2), a piledriver hammer shall be furnished. The piledriver hammer shall conform to MIL-H-21103, size as specified.

3.18.2.10 Shovel-front. When specified (see 6.2), a shovel front shall be furnished, and shall operate in accordance with the requirements of table H.

3.18.2.10.1 Shovel boom. The shovel boom shall be of sturdy construction, strongly reinforced and braced, and shall be fabricated from structural steel shapes and plates rigidly welded to form a box section up to the dipper shaft for an inside dipper stick and to the boom tip for an outside dipper stick. Replaceable wearing shoes or other suitable fittings shall be provided where the dipper stick passes by or through the boom. Removable and replaceable buffer rails shall be provided where the dipper strikes the boom.

3.18.2.10.2 Dipper stick. The dipper stick shall be of rigid and durable steel construction and shall resist twisting strains.

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3.18.2.10.3 Shovel dipper. The shovel dipper shall be of either heavy steel plate, cast steel, or a combination of both. The front lip shall be of abrasion- and wear-resistant, hard-surfaced steel. The bucket lips shall be equipped with tooth sockets, shanks, and teeth in accordance with MIL-T-52176. The dipper shall have a discharge gate, greater in area than the dipper mouth, in the bottom, securely hinged to the back of the dipper with large steel hinges. The dipper front-latch opening on the wearing side shall have a replaceable wear plate where the latch bar engages. The dipper angle shall be adjustable for different digging conditions. A power trip controlled from the operator's seat and a slack take-up to prevent fouling of the trip line shall be provided. The shovel dipper capacity shall be not less than 2 cubic yards, obtained by multiplying the mean height of the dipper by the inside cross-sectional area of the dipper at one-half the minimum height ± 2 percent. The mean height shall be one-half the sum of the maximum and minimum height of the dipper body including its minimum lip projection without teeth.

3.18.2,10.4 Dipper crowd and retract. The dipper crowd shall be self-adjusting at all boom angles and shall be provided with a tension adjustment for maintaining the correct tension in the chain or rope. The dipper retract and crowd shall be operated by an independent positive chain or wire rope mechanism.

3.18.2.10.5 Sheave. The shovel boom point sheaves shall conform to 3.15.7 except that the pitch diameter shall be not less than 32 times the diameter of the rope used.

3.18.2.11 Backhoe. When specified (see 6.2), a backhoe shall be furnished and shall operate in accordance with table 111.

3.18.2.11.1 Boom, dipper arm, and gantry. The boom, dipper arm, and gantry shall be made of rigidly welded or riveted steel members.

3.18.2.11.2 Dipper. The backhoe dipper shall be constructed of heavy steel plate or of cast steel; shall have a cutting lip of abrasion- and wear-resistant hard-surfaced steel; shall have the bottom reinforced and protected by the extension of the tooth sockets or adequate wearing bars; shall have teeth in accordance with MIL-T-52176, and shall have replaceable, abrasion and wear-resistant, heat treated and tempered, drop-forged steel, or cast alloy steel side cutter.

3.19 Lubricants. The procedure for the selection of lubricants shall be in accordance with MIL-STD-838, section 5. Lubricants selected shall be in accordance with MIL-HDBK-113, chapter 2. When the specification of the lubricant selected includes a requirement for a Qualified Products List (QPL), the lubricant supplied shall be from a source that is listed on the applicable QPL.

3.19.1 Lubrication. All surfaces requiring lubrication shall be provided with means of lubricating. All parts shall be lubricated with military lubricants unless it is a permanently lubricated part with sealed lubricant and no lubricant

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change shall be required in the field. A lubrication data plate of permanent construction shall be mounted on each crane. The lubrication data plate shall conform to 3.20.1.

3.19.2 Lubrication fittings. Fittings in accordance with SAE J534 shall be located in protected positions, and shall be accessible to, and compatible with, a grease gun conforming to MIL-G-3859 equipped with a hydraulic coupler conforming to MS24207 and a lo-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts, except that engine-housing side panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed.

3.20 Identification marking. The crane markings shall conform to MIL-STD-130. The marking shall be applied on plates conforming to MIL-P-514, type I, style I, composition C of type I, grade A, class I material. Plates shall be attached by screws, bolts or rivets in a location where the plate will be both visible and legible.

3.20.1 Instruction plates or labels. Each major component of the crane shall be equipped with instruction plates, or labels, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating or servicing the crane. Instruction plates and labels shall conform to MIL-P-514, type III, composition C of type I, grade A, class 1 material and the labeling requirements of 3.7.2. Plates or labels shall be attached by screws, bolts, or rivets in a location where the plate or label shall be both visible and legible. Required danger and caution signs shall conform to MIL-P-514.

3.20.2 Shipping data plate. A shipping data plate conforming to MIL-P-514, type III, composition C of type I, grade A, class 1 material shall be provided. The plate shall show a silhouette of the crane in transport configuration indicating the center of gravity, the location and capacity of the slings and tiedown provisions. The plates shall be attached by screw, bolts or rivets in a location where the plate shall be visible and legible. A plate shall be provided on each crane.

3.21 Treatment and painting. The portions of the crane normally painted shall be cleaned, treated and painted in accordance with MIL-T-704, type F or G, as applicable. Unless otherwise specified (see 6.2), top coat shall be camouflage green 383 conforming to MIL-C-53039. Surfaces to be painted that exceed 400 *F shall be painted with paint conforming to MIL-P-14105 in a color matching that specified.

3.22 Tool/storage box. A tool/storage box shall be provided on each crane. The box shall be of sufficient size to accommodate the maintenance and special tools as required to perform the routine and operator maintenance, and all nuts, bolts, and small components removed for transport. The box shall be welded steel construction. The box cover shall be attached to box with a continuous hinge on one side and the cover shall extend down over the sides of the box at least 5/8 inch. The box cover shall be secured utilizing a padlock eye minimum size 0.625 inches. A keyed padlock of suitable size shall be provided. The box shall

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be located in an accessible, protected area. Means for drainage of the box shall be provided. The box opening shall be located for accessibility.

3.23 Government-furnished equipment (GFE). When specified (see 6.2), the following property will be furnished by the Government (see 6.5):

Item No.	Description	'Identification	Quantity
1	Bracket, mounting decontaminating apparatus, portable (DAP) , M13	MS53052-1	As required
2	Bracket assembly, rifle stowage	Drawing 13229E8014	As required

3.24 Government-loaned propertv. Unless otherwise specified (see 6.2), the following equipment in the quantities indicated will be loaned by the Government (see 6.6):

Item No.	Description	Identification (NSN)	Quantity
1.	Lead section, top, pile driver: 15 ft. long.	3815-00-983-8029	1
2.	Lead section, lower, pile driver: 10 ft. long.	3815-00-221-2215	4
3.	Catwalk, pile driver, telescoping: 3 sect; 8 to 23 ft.	3815-00-190-3308	1
4.	Hammer, pile driver, self-powered: diesel driven, 8,000 ft. lb., size D	3895-00-769-5308	1

3.25 Servicing and adjustment. The contractor shall service and adjust the cranes for operational use to include at least the following: Adjustment of the lights and brake systems, adjustment of mechanical and electrical drive components, including drive chains and belts, adjust and align all belts, and clean and fill all hydraulic units. The cranes shall be completely lubricated as specified in 3.19.1.

3.26 Workmanship.

3.26.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame cutting, using tips suitable for the thickness of the metal, may be used instead of shearing and sawing. All bends shall be made by controlled means to ensure uniformity of size and shape. Precautions shall be taken to avoid overheating. Heated metal shall be allowed to cool slowly. External surfaces shall be free from burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

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3.26.2 Bolted connections. Bolt holes shall be accurately formed and shall have the burrs removed. Washers and lock washers shall be provided where necessary. Matching threaded areas fastened by bolts conforming to SAE J429 or by cap screws conforming to SAE J478 shall be of sufficient strength to withstand the tensile strength of t-he bolt or capscrew. All bolts shall be correctly torqued and shall have full thread engagement. Bolts and capscrews shall protrude not less than two and no more than five full threads beyond the nut.

3.26.3 Welders and welding operators. Before assigning any welder or welding operator to manual welding work covered by this specification, the contractor shall obtain certification that the welder or welding operator has passed qualification tests as prescribed by either AWS D1.1, 1114.3, or the ASME codes for the materials joined and the type of welding operation to be performed and that such qualification is effective as defined by the particular code. Contractors who only make horizontal welds need not qualify welders for “all position welding”. The contractor is responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS and ASME codes. In the event of evidence of poor welds, the Government reserves the right to require retesting of any welder or welding operator. The test results shall be made available for review by the contracting officer or the contracting officer’s representative (see 6.2.1).

3.26.4 Welding. Welding procedures and welds shall be in accordance with ASME or AWS codes. The surfaces of parts to be welded shall be free from rust, scale, paint, grease and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are as specified herein. All parts, components, and assemblies of the crane shall be cleaned and free of any harmful extraneous material. External surfaces shall be free of burrs, sharp edges, and corners except when sharp edges or corners are required.

3.26.5 Castings and forging. All castings and forgings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework shall not reduce typical mechanical properties or affect a castings function.

3.26.6 Riveted connections. Rivet holes shall be accurately formed or drilled and shall have the burrs removed. Rivets shall completely fill the holes. Rivet heads shall be concentric with the rivet, and shall make full contact with the intended surface. The rivets shall be so fabricated as to develop a joint strength not less than the design value.

3.27 Decontaminating.j apparatus, portable (DAP), M13. When specified (see 6.2), a GFE DAP mounting bracket (MS53052-1) shall be installed in a manner which shall not Interfere with the normal operation of the equipment, is accessible to personnel, and provides adequate clearance to remove and emplace a DAP M13 unit having a maximum weight of 54 pounds. An upright installation is required. The DAP device has a flash point of 165 ^{0}F and shall be installed away from exhaust or heat sources. The DAP mounting bracket shall be mounted within or near the operator cab (see 6.2).

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3.2$ Bracket assembly, rifle stowage. When specified (see 6.2), GFE rifle stowage bracket assemblies conforming to Drawing 13229E8014, shall be furnished and mounted by the contractor. The location of-the bracket shall be such that it is accessible to the operator and provides clearance for the rifle to be removed. The bracket shall be located away from operators's controls and shall not impair the vision of the operator. Bracket assemblies shall be mounted within or near the operator cab (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, facilities suitable for the performance of the inspection requirements specified herein may be the contractor's, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Examination. Each crane shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements, shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirement or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.2.1 Component and material inspection. The contractor is responsible for ensuring that components and materials are manufactured, examined and tested in accordance with referenced specifications and standards, as applicable.

4.3 First article inspection.

4.3.1 First article examination. As specified in 4.5.1, the first article shall be examined for the defects listed in 4.5.1. The presence of one or more defects shall be cause for rejection.

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4.3.2 First article tests. The first article shall be tested as specified in 4.5.2. AN first article tests shall be run using JP-8. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. As specified in 4.5.1, each crane shall be examined for the defects listed in table V. The presence of one or more defects shall be cause for rejection.

4.4.2 Tests. As specified in 4.5.2.1, each crane shall be subjected to the tests marked "X" in the quality conformance column of table VI. ‘These tests may be run using either JP-8 or diesel fuel. Failure of any test shall be cause for rejection.

4.5 Inspection procedures. Prior to examination and tests, the crane shall be set-up with all auxiliary equipment and accessories installed. Each crane shall be Lubricated with military or military compatible oil and grease for use in the ambient temperature range present at the inspection site. As the services and adjustments are performed, the crane shall be observed for compatibility between accessories and ease of maintenance.

4.5.1 Examination. The crane shall be examined as specified herein for the defects listed in table V:

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TABLE V. Examination schedule.

Number	Defect	Requirement Paragraph
101.	Crane not as specified.	3.1
102.	Materials not as specified.	3.4
103.	Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration of the applicable storage and operating environments.	3.4.1
104.	Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	3.4.2
105.	Cadmium plating used when not necessary.	3.4.3
106.	Used, rebuilt or remanufactured components, pieces, or parts incorporated in the crane.	3.4.4
107.	Safety not as specified.	3.6.1
108.	Human engineering not as specified.	3.7
109.	Noise limits not as specified.	3.8
110.	Transportability not as specified.	3.9
111.	Disassembly and reassembly not as specified.	3.9.1
112.	Tiedown/Slinging provisions not as specified.	3.9.2, 3.9.3
113.	Rail transportability not as specified.	3.9.4
114.	Highway tranportability not as specified.	3.9.5
115.	Antirotation device not as specified.	3.9.6
116.	Maintainability not as specified.	3.10
117.	Filling, draining, and checking provisions not as specified.	3.10.1
118.	Operating temperature not as specified.	3.11.1
119.	Operations at elevations not as specified.	3.11.2
120.	Lifting capacities not as specified.	3.12.1.1
121.	Shovel capacities not as specified.	3.12.1.2
122.	Backhoe capacities not specified.	3.12.1.3

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TABLE V. Examination schedule. (continued)

Number	Defect	Requirement Paragraph
123.	Clamshell and dragline capacities not as specified.	3.12.1.4
124.	Crane capacity plate not as specified.	3.12.1.5
125.	Attachment operations not as specified.	3.12.2
126.	Maneuverability not as specified.	3.12.2.1
127.	Ground bearing pressure not as specified	3.12.2.2
128.	Backward stability not as specified.	3.13
129.	Load stability not as specified.	3.14
130.	Stress requirements not as specified.	3.15
131.	Superstructure not as specified.	3.16, 3.16.1, 3.16.2
132.	Cab not as specified.	3.16.3, 3.16.3.1, 3.16.3.2
133.	A-frame not as specified.	3.16.4
134.	Power load not as specified.	3.16.5
135.	Crane-boom hoist not as specified.	3.16.6
136.	Drums and sheaves not as specified.	3.16.7
137.	Drum brakes not as specified.	3.16.7.1
138.	Drum clutches not as specified.	3.16.7.2
139.	Auxiliary drum not as specified.	3.16.7.3
140.	Rollers not as specified.	3.16.8
141.	Swing mechanism not as specified.	3.16.9
142.	Swing gear, lock, and brake not as specified.	3.16.9.1
143.	Power transmission not as specified.	3.16.10
144.	Power transmission clutch not as specified.	3.16.10.1
145.	Power transmission drive chains not as specified.	3.16.10.2
146.	Electrical system not as specified.	3.16.11
147.	EMI suppression not as specified.	3.16.11.1

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TABLE V. Examination schedule. (continued)

Number	Defect	Requirement Paragraph
148.	Wiring not as specified.	3.16.11.2
149.	Lights not as specified.	3.16.11.3
150.	Floodlights not as specified.	3.16.11.3.1
151.	Interior lights not as specified.	3.16.11.3.2
152.	Trouble lamp not as specified.	3.16.11.3.3
153.	Warning horn not as specified.	3.16.11.4
154.	Startting systems not as specified.	3.16.11.5
155.	Charging systems not as specified:	3.16.11.6
156.	Fungus and moisture resistance not as specified.	3.16.11.7
157.	Batteries not as specified.	3.16.11.8
158.	Battery enclosures not as specified.	3.16.11.9
159.	Diesel engine not as specified.	3.16.12
160.	Oil sampling valve not as specified.	3.16.12.1
161.	Diagnostic connector assembly (DCA) not as specified.	3.16.12.2
162.	Operational conditions not as specified.	3.16.12.3
163.	Air cleaner not as specified.	3.16.12.4
164.	Engine exhaust system not as specified.	3.16.12.5
165.	Governor not as specified.	3.16.12.6
166.	Operator controls not as specified.	3.16.13
167.	Crawler base not as specified.	3.17.1
168.	Track assembly not as specified.	3.17.2
169.	Side frames not as specified.	3.17.3
170.	Crawler drive not as specified.	3.17.4
171.	Axles not as specified.	3.17.5
172.	Front-end attachments not as specified.	3.18
173.	Crane boom not as specified.	3.18.1.1

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TABLE V. Examination schedule. (continued)

Number	Defect	Requirement Paragraph
174.	Pendant equalizer assembly not as specified.	3.18.1.2
175.	Pendant cable not as specified.	3.18.1.3
176.	Boom safety stop not as specified.	3.18.1.4
177.	Boom hoist clutch throwout not as specified.	3.18.1.5
178.	Boom angle indicator not as specified.	3.18.1.6
179.	Overload indicating system not as specified.	3.18.1.7
180.	Crane hook block not as specified.	3.18.1.8
181.	Wire rope not as specified.	3.18.1.9
182.	Rope sockets not as specified.	3.18.1.9.1
183.	Rope lengths not as specified.	3.18.1.9.2 and table IV
184.	Boom extensions not as specified.	3.18.2.1
185.	Boom jib not as specified.	3.18.2.2
186.	Clamshell bucket not as specified.	3.18.2.3
187.	Tagline not as specified.	3.18.2.4
188.	Dragline bucket not as specified.	3.18.2.5
189.	Dragline fairlead not as specified.	3.18.2.6
190.	Wrecking ball not as specified.	3.18.2.7
191.	Piledriver leads and catwalks not as specified.	3.18.2.8
192.	Piledriver hammer not as specified.	3.18.2.9
193.	Shovel-front not as specified.	3.18.2.10
194.	Shovel boom not as specified.	3.18.2.10.1
195.	Dipepr stick nto as specified.	3.18.2.10.2
196.	Shovel dipper not as specified.	3.18.2.10.3
197.	Dipper crowd and retract not as specified.	3.18.2.10.4
198.	Boom point sheaves not as specified.	3.18.2.10.5

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TABLE V. Examination schedule. (continued)

Number	Defect	Requirement Paragraph
199.	Backhoe not as specified.	3.18.2.11 and table III
101a.	Boom dipper arm, and gantry not as specified.	3.18.2.11.1
102a.	Dipper not as specified.	3.18.2.11.2
103a.	Lubricants not as specified.	3.19
104a.	Lubrication not as specified.	3.19.1
105a.	Lubrication fittings not as specified.	3.19.2
106a.	Identification marking not as specified.	3.20
107a.	Instruction plates or lables not as specified.	3.20.1
108a.	Shipping data plate not as specified.	3.20.2
109a.	Treatment and painting not as specified.	3.21
110a.	Tool/storage box not as specified.	3.22
111a.	Servicing and adjustment not as specified.	3.24
112a.	Metal fabrication not as specified.	3.26.1
113a.	Bolted connections not as specified.	3.26.2
114a.	Welders and welding operators not as specified.	3.26.3
115a.	Welding not as specified.	3.26.4
116a.	Castings and forgings not as specified.	3.26.5
117a.	Riveted connections not as specified.	3.26.6
118a.	Decontamination apparatus (DAP) not as specified.	3.27
119a.	Rifle stowage bracket not as specified.	3.28

4.5.2 Tests.

4.5.2.1 Test schedule.

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TABLE VI. Test schedule.

First Article	Quality Conformance	Test	Test Paragraph	*Requirement Paragraph
X	X	Operational	4.5.2.2	3.1, 3.11, 3.12, 3.15, 3.16, 3.17, 3.18, 3.19, 3.25
X	X	Maneuverability	4.5.2.3	3.12.2.1
X	X	Ground bearing	4.5.2.4	3.12.2.2
X	X	Stability	4.5.2.5	3.13, 3.14
X	X	Lifting capacity	4.5.2.6	3.12.1.1, 3.15
X	X	Swing	4.5.2.7	3.16.1, 3.16.8, 3.16.9, 3.16.9.1
X	X	Travel brake and lock	4.5.2.8	3.17.4
X	X	Powered load lowering	4.5.2.9	3.16.5
X	X	Load holding	4.5.2.10	3.16.7.1
X	X	Pile driving	4.5.2.11	3.1, 3.18.1.9.2, 3.18.2.8, 3.18.2.9
X	X	Backhoe	4.5.2.12	3.12.1.3
X	X	Shovel-front	4.5.2.13	3.12.1.2
X	X	Wrecking	4.5.2.14	3.18.2.7
X	X	Clamshell and dragline	4.5.2.15	3.18.2.3, 3.18.2.4, 3.18.2.5, 3.18.2.6
X	X	EMI	4.5.2.16	3.16.11.1
X	X	Noise	4.5.2.18	8.7
X		Transportability	4.5.2.19	3.9, 3.9.1, 3.9.4, 3.16.3.2
X		Tiedown and slinging provisions	4.5.2.20	3.9.2, 3.9.3
X		Rail transport	4.5.2.21	3.9.4

* Paragraphs listed include all subparagraphs.

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4.5.2,2 Operational test. Prior to delivery, the complete crane shall be assembled, lubricated, adjusted, and prepared for operation. Operate the complete crane including all components and accessories for a period of 30 minutes. Operation shall include demonstration of controls and speeds specified herein. Any malfunction, overheating, leakage, or failure to complete the test for any reason shall be cause for rejection. If such failures or deficiencies do occur, corrections shall be made and the operational test repeated. Prior to disassembly for shipment or transport, all accessories, fittings, piping, railings, connections, etc., shall be provided with painted match marks for easy identification when being reassembled at destination. Inability to meet the operational requirements of 3.1, 3.11, 3.14, 3.15, 3.16, 3.17, 3.18, 3.19, and 3.25 shall constitute failure of this test.

4.5.2.3 Maneuverability test. The crane shall be tested to ensure that its maneuverability conforms to that specified in 3.12.2.1.

4.5.2.4 Ground bearing pressure test. The ground bearing pressure shall be calculated by dividing the total working weight with basic 40-foot boom, but not including load, by the bearing area in square inches. The weight shall be determined by platform scales, by calibrated hydraulic, or electronic devices and the bearing area used shall be the center-to-center distance between crawler end sprockets multiplied by twice the width of the track shoe.

4.5.2.5 Stability test. Crane stability test shall be conducted in accordance with the procedures of SAE J765. Inability to meet the requirements of 3.13 and 3.14 shall constitute failure of this test.

4.5.2.6 Lifting capacity tests. Lifting capacities of crane shall be demonstrated by use of actual weights using the minimum (40-foot), maximum (160-foot), and one intermediate length boom. The loads at the tipping point and the boom radius shall be measured as specified in SAE J765 for 3 radii at each boom length. Inability to meet the requirements of 3.12.1.1 and 3.15 shall constitute failure of this test.

4.5.2.7 Swing test. A swing test shall be conducted to demonstrate the following requirements. Inability to meet these requirements and those of 3.16.1, 3.16.8, 3.16.9, and 3.16.9.1 shall constitute failure of this test.

a. With the 40-foot boom at minimum radius, rotate the superstructure at full throttle for 10 minutes in each direction in order to determine if excessive heat is generated in the rollers in conformance to 3.16.8.

b. With the 40-foot boom at minimum radius, continually oscillate the superstructure at full throttle through an arc of 180 degrees for 10 minutes, to determine if excessive heat is generated in swing clutches.

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c. Test the swing lock or brake at an engine speed of not less than 200 rpm above low idle speed by actuating it in several positions and attempting to swing the superstructure from those positions.

d. Test for a swing speed conforming to 3.16.1.

4.5.2.8 Travel brake and lock test. Test the service brakes or other locking means to demonstrate conformance to 3.17.4. Inability to meet the requirements of 3.17.4 shall constitute failure of this test.

4.5.2.9 Powered load lowering test for crane. The powered load lowering system shall be tested as follows. Inability to meet the requirements of 3.16.5 shall constitute failure of this test.

a. With the crane equipped with boom, position the revolving superstructure in the most stable craning position and assemble a single line load equal to the maximum rated load-divided by the recommended number of parts of hoist line reeving.

b. Position the boom angle for the minimum working radius.

c. Hoist the load 20 feet above the ground level, ± 3 inches, or until interference with the crane boom is obvious.

d. Prepare the crane for power-down operation on the main load line.

e. Lower the load under power. Vary the speed of the load by varying the speed of the engine.

f. When the load reaches the ground, immediately repeat the operations c, d, and e. These operations shall be repeated until 20 cycles have been completed.

g* Check the temperatures of all linings employed in this operation with at least four thermocouples imbedded within each lining, near the working face, and uniformly spaced around the perimeter, or with a contact pyrometer placed directly on the lining, or on the drum surface near where the lining contacts the drum.

4.5.2.10 Load holding test. Suspend load; stop the engine; positively lock the main drive mechanism, positively apply the primary hoist drum clutch and release the brake holding the load. As the brake is released, inspect the clutch for slippage. The clutch shall have adequate capacity to hold this load without slipping. Repeat this test using the secondary hoist drum clutch. Suspend a capacity load; stop the engine; positively engage primary drum clutches; disengage the master clutch and lower the load on the brake, backing down the hoist machinery without raising the boom. Repeat this test using the secondary drum clutch. An inspection shall be made for ease of control operation, smoothness of load lowering, and machinery operation. This operation shall be performed without

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binding, chattering of machinery, slipping at the clutches, or unusual effort of the operator. Inability to meet the requirements of 3.16.7.1 shall constitute failure of this test.

4.5.2.11 Pile driving test. When applicable (see 6.2), attach the Government loaned pile driving leads, catwalk, and hammer to the boom and revolving superstructure. Perform pile driving operations for 4 hours. Inability to operate with pile driving hammer, malfunction, evidence of any damage, and failure of any component or to perform the requirements of 3.1, 3.18.1.9.2, 3.18.2.8, and 3.18.2.9 shall constitute failure of this test.

4.5.2.12 Backhoe test. When furnished with a backhoe attachment (see 6.2), perform maximum capacity heavy-duty trenching operations in undisturbed soil including digging depth specified in table III for not less than 4 hours. Failure to perform as specified in 3.12.1.3, any evidence of malfunction, damage, or permanent deformation, shall constitute failure of this test.

4.5.2.13 Shovel-front attachment test. When furnished with a shovel-front attachment (see 6.2) perform shovel-front excavating operations at maximum capacity, and include equal portions of time operating at each of the ranges specified in table II. The total period of operation shall be not less than 4 hours. The operations shall be conducted in a gravel pit or rock quarry. Failure to perform as specified in 3.12.1.2, any evidence of malfunction, damage, or permanent deformation shall constitute failure of this test.

4.5.2.14 Wrecking test. Perform simulated wrecking operations both in the horizontal and vertical planes for a total of 4 hours. Failure to perform as specified in 3.18.2.7, inability to operate with wrecking ball, malfunction, evidence of any damage, or failure of any component shall constitute failure of this test.

4.5.2.15 Clamshell and dragline test. Perform maximum-capacity, heavy-duty clamshell and dragline operations in undisturbed heavy clay soil. The operation shall be performed with the clamshell and dragline bucket for at least 8 hours (4 hours for each bucket). Inability to operate with clamshell and dragline buckets as specified in 3.18.2.3, 3.18.2.4, 3.18.2.5, and 3.18.2.6, inability to complete the test, malfunction, evidence of any damage, or failure of any component shall constitute failure of this test.

4.5.2.16 Measurement of electromagnetic radiation. The electromagnetic radiation shall be measured in accordance with SAE J551. Inability to meet the requirements of 3.16.11.1 shall constitute failure of this test.

4.5.2.17 Human engineering. The crane shall be operated and maintenance actions shall be performed (either by actual performance or by simulation) and the design shall be reviewed and analyzed to determine conformance to 3.7. Nonconformance to 3.7 shall constitute failure of this test.

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4.5.2.18 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 11. As a minimum, noise levels shall be measured when equipment is operating under full load. MIL-STD-1474, 5.1.2.1.4 contours shall be taken at not fewer than 12 equal horizontal arc increments; one increment shall include data from the noisiest position. Additionally, the noise level at the typical operating position shall be provided as dB(A) level. Failure to comply with MIL-STD-1474 and 3.8 provisions shall constitute failure of this test.

4.5.2.19 Transportability. The crane and components shall be weighed, measured, and analyzed to determine conformance to 3.9, 3.9.1, 3.9.6, and 3.16.3.2.

4.5.2.20 Tiedown and slinging test. The tiedown and slinging provisions shall be tested in accordance with MIL-STD-209 to prove conformance with 3.9.2 and 3.9.3. Inability to meet these requirements shall constitute failure of this test.

4.5.2.21 Rail transportability test. The first article crane shall be tested in accordance with MIL-STD-180 to prove conformance to 3.9.4. Inability to meet the requirements of 3.9.4 shall constitute failure of this test. .

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a complete pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Examination. Each unit of product shall be examined for the following defects. The presence of one or more defects shall be cause for rejection.

124a. Disassembly and match marking not as specified (see 5.1.1 1.1).
125a. Methods of preservation not as specified (see .5.1.1.1.2).
126a. Cleaning and drying not as specified (see 5.1.1.1.3).
127a. Unprotected and unfinished exterior metal surfaces not preserved as specified (see 5.1.1.1.4 and 5.1.1.1.4.1).
128a. Machined surfaces not preserved as specified (see 5.1.1.1.4.2).
129a. Gears not preserved as specified (see 5.1.1.1.5 thru 5.1.1.1.5.2).
130a. Drive chains not preserved as specified (see 5.1.l.l.6 thru 5.1.1.1.6.2).
131a. Brake systems not preserved as specified (see 5.1.1.1.7 thru 5.1.1.7.2).
132a. Hydraulic systems not preserved as specified (see 5.1.1.1.8).
133a. Belts and pulleys not preserved as specified (see 5.1.1.1.9).
134a. Pumps not preserved as specified (see 5.1.1.1.10).
135a. Fuel tanks not preserved as specified (see 5.1.1.1.11).
136a. Electrical components not preserved as specified (see 5.1.1.1.12).
137a. Openings not sealed to prevent entry of water (see 5.1.1.1.13).
138a. Loose piping equipment not preserved as specified and exposed thread not protected against damage as specified (see 5.1.1.1.14).

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139a. Grease fittings, wheel bearings, control mechanisms, linkage, hinges, fasteners, and pivot points not lubricated as specified (see 5.1.1.1.15).
140a. Maintenance tools not preserved as specified (see 5.1.1.1.16).
141a. Technical publications not preserved as specified (see 5.1.1.1.17).
142a. Consolidation not as specified (see 5.1.1.1.18).
143a. The complete crane not preserved in a manner to prevent deterioration and damage (see 5.1.2).
144a. Components not packed as specified (see 5.2.1.1).
145a. Piping and accessories not bundled or packed as specified (see 5.2.1.2).
146a. Cranes not packed as specified for level C (see 5.2.2).
147a. Marking missing, illegible, or incorrect (see 5.3).

5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article pack for examination within the time frame specified (see 6.2), to prove, prior to-starting production packaging, that the applied preservation, packing and marking comply with the requirements of this purchase description. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government. The first article pack may be accomplished utilizing either the first article model or a production model. If the first article model is. utilized, and the Government requests a comparison between the first article model and a production model, any preservation and packing shall be removed by the contractor at no expense to the Government.

5.1.1 Preservation. The preservation shall be level A or C as specified (see 6.2).

5.1.1.1 Level A.

5.1.1.1.1 Disassembly and match marking. Disassembly shall be the minimum necessary to safeguard parts known to be subject to damage or loss, and to accomplish reduction in cube. Bolts, nuts, screws, pins and washers removed shall be reinstalled in one of the mating parts and secured to prevent their loss. When necessary to facilitate reassembly, parts removed and mating parts shall be match marked. Large sections and components shall be match marked with lusterless white enamel letters. Match marking shall be over-coated with clear spar varnish conforming to TT-V-121. Small parts that are removed shall have match marking information marked on weatherproof tags attached to mating parts. Waterproof ink shall be used for marking of information on tags.

5.1.1.1.2 Methods of preservation. Cleaning processes, drying procedures, preservatives and methods of preservation are listed in MIL-P-116 and shall conform to the requirements of MIL-P-116 and any applicable specifications.

5.1.1.1.3 Cleaning and drying. Prior to the application of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

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5.1.1.1.4 Unprotected surfaces. Unprotected exterior metal surfaces requiring the application of a contact preservative conforming to MIL-P-116 and not specifically provided for herein shall be preserved as follows:

5.1.1.1.4.1 Unfinished (not machined) SUrfaces. Unfinished exterior metal surfaces shall be coated with type P-1 preservative.

5.1.1.1.4.2 Machined surfaces. Exposed machined surfaces, shall be coated with type P-6 or P-11 preservative and wrapped or covered, as applicable, with barrier material conforming to MIL-B-121, type I, grade A, class 2 or MIL-B-22191, type H. The material shall be secured in place with waterproof tape.

5.1.1.1.5 Gears.

5.1.1.1.5.1 Exposed gears. Exposed gears shall be coated with type P-1 preservative, or a thin film of primer conforming to MIL-P-62669.

5.1.1.1.5.2 Enclosed sears. Gear housings containing lubricant conforming to MIL-L-2105 shall be checked for operating level and the gears actuated to ensure coating of all interior surfaces.

5.1.1.1.6 Drive chains.

5.1.1.1.6.1 Enclosed chains. Enclosed chains and chain housings shall be preserved as specified for enclosed gears.

5.1.1.1.6.2 Exposed chains. Exposed chains shall be coated with enough P-3 or P-9 preservative to assure penetration of the preservative to the inner surfaces of the rollers, pins and bushings. After the excess preservative has drained, the chain and sprockets shall be coated with type P-1 preservative.

5.1.1.1.7 Brakes.

5.1.1.1.7.1 Brake drum assemblies. All metal surfaces enclosed within brake drums and the braking surface or face of the brake drums shall be coated with a thin film of primer conforming to MIL-P-62669. Care shall be exercised to prevent the primer from coming in contact with brake lining, rubber and rubber impregnated parts.

5.1.1.1.7.2 Air actuated brake systems. The air actuated brake systems shall be drained of condensation. Interior surfaces of the air supply tanks shall be fogged with type P-10 preservative. Excess preservative shall be allowed to drain and the drain plugs reinstalled after preserving the threads with P-10. Air line filters shall be drained and closed. The exhaust ports of relay emergency quick release, and relay valves not equipped with exhaust check valves, shall be closed by inserting pipe plugs or by sealing the ports with waterproof tape. Weatherproof tags shall be attached to the valves and service line indicating that: 'The plugs or tape must be removed from the exhaust ports before placing the equipment in operation." Waterproof ink shall be used for marking of information on tags.

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5.1.1.1.8 Hydraulic systems. The hydraulic fluid supply tank shall be filled to the operating level with the operation fluid (see 3.18) to revent moisture ingression. Where practicable, pistons shall be in their fully retracted position, with any remaining unpainted surfaces coated with type P-1 preservative. Where pistons cannot be fully retracted, exposed areas or normally retractable surfaces shall be coated with type P-6 or P-11 preservative. The type P-6 or P-n preservative coated surfaces shall be spirally wrapped with barrier material conforming to MIL-B-121, type II, grade A, class 2. The wraps shall be secured in place with waterproof tape. All other unpainted exterior surfaces shall be coated with type P-1 preservative. All assemblies actuated by the hydraulic system shall be secured.

5.1.1.1.9 V-belts. flat belts. and pulleys. Unpainted surfaces of pulleys shall be coated with primer, conforming to MIL-P-62669. The belts for each crane shall be preserved method IC-1 or IC-3. Large flat belts shall be preserved and packaged as specified for conveyor belts.

5.1.1.1.10 Pumps. The interior surfaces of the pump, piping, valves, jets and nozzles shall be coated with type P-10 preservative in a manner to ensure thorough coating of all interior surfaces. Excess preservative shall be drained. All openings shall be sealed with metal or plastic caps or plugs, or sealed with waterproof tape.

5.1.1.1.11 Fuel tanks. The interior surfaces of fuel tanks, if not coated or not fabricated of corrosion-resistant material, shall be fogged with type P-10 preservative. Excess preservative shall be drained and the threads of drain plugs shall be coated with P-10 before reinstalling.

5.1.1.1.12 Electrical components. Electrical motors shall be preserved level A by the alternate requirements of MIL-E-16298. Other electrical components shall be preserved conforming to MIL-E-17555 level A.

5.1.1.1.13 Openings. Openings in the equipment which will trap free falling water shall be sealed by any suitable means to prevent the entry of water.

5.1.1.1.14 Piping equipment. Loose piping equipment required to complete the water-supply system shall be preserved in accordance with MIL-P-116, method I or III, as applicable. Exposed threaded surfaces shall be protected against damage by the use of metal or plastic thread protectors.

5.1.1.1.15 Lubrication. Lubricate all grease fittings with type P-11 preservative. Wheel bearing should be lubricated with MIL-G-10924, GAA, grease. Lubricate control mechanisms, linkage, hinges, fasteners and all pivot points with P-lo.

5.1.1.1.16 Maintenance tools. Maintenance tools shall be preserved conforming with level A preservation and packaging requirements of PPP-P-40.

5.1.1.1.17 Technical publications. Technical publications for each piece of equipment shall be preserved conforming to MIL-P-116, method IC-1 or IC-3.

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9 5.1.1.1.18 Consolidation. The tool boxes shall be utilized for' the consolidation of loose components, tools and publications. The remainder of the components that will not fit in the tool boxes shall be consolidated, along with disassembled components, in boxes conforming to PPP-B-601, overseas type or PPP-B-621, ClaSS 2. Boxes shall be placed in the cab or secured to the equipment with strapping conforming to ASTM D 3953, type 1 or 2, zinc-coated, size as applicable and ASTM D 4675. Arrangement and location on the equipment shall not increase cubical volume or interfere with lifting or mobility. of the equipment.

5.1.2 Level C. The complete crane shall be preserved and packed in accordance with ASTM D 3951.

5.2 Packing. The packing shall be level A or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Components. Components shall be packed in boxes conforming to PPP-B-601, overseas type, or PPP-B-621, class 2. Components shall be securely blocked, braced or cushioned within the boxes to prevent movement. When sufficient space is available, packaged and un-packaged components required for each crane or section shall be placed and secured on the crane or section for which intended in any available space that shall not interfere with lifting or mobility of the cranes. Boxed and crated components shall be shipped at the same time as the cranes. Packing and containers shall comply with applicable carrier rules and regulations. The toolbox lids shall be closed and secured in a manner to prevent pilferage.

a– 5.2.1.2 Piping and accessories. Piping suitable for bundling shall be bundled in secure lifts. The piping shall be nested, arranged and secured with bolts or steel straps, or a combination of both, to form compact non-shifting bundles. Suitable wood blocking shall be used, as required, under strapping to eliminate large voids or irregular shaped bundles. Strapping shall be 0.023 by 0.75 inch flat steel strapping conforming to ASTM D 3953, type 1 or 2, zinc-coated and ASTM 5 4675. Strapping shall be spaced not to exceed approximately 36 inches on center, with end straps not more than 18 inches from each end. Metal edge protectors shall be used under strapping when strapping bears on the pipe. The bundles shall be packed in open type crates conforming to MIL-C-52950, style B, type III. Short pieces of pipe, hose fittings and accessories shall be packed in close-fitting boxes conforming to PPP-B-621, class 2 or PPP-B-601, overseas type. The contents in the boxes and crates shall be cushioned, blocked and braced.

5.2.2. Level C. The complete crane shall be prepared for shipment in a manner which will ensure arrival at destination in satisfactory condition. Loose components that are subject to damage during shipment shall be blocked, braced, or otherwise secured to prevent damage during shipment. The toolbox lids shall be closed and secured in a manner to prevent pilferage. When sufficient space is available, packaged and un-packaged components required for each crane or section shall be placed and secured on the crane or sections for which intended in any available space that shall not interfere with lifting or mobility of the cranes.

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Boxed and crated components shall be shipped at the same time as the cranes. Packing and containers shall comply with applicable carrier rules and regulations.

5.3 Marking. In addition to any special marking required (see 6.2), marking for shipment and storage shall conform to MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The crane, together with the various items of front-end operating equipment, are intended for use in handling materials, lifting and transporting loads short distances, soil stripping, earthmoving, ditch digging, pile driving, and demolition.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue-of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is not required (see 3.3).
d. When the shovel-front attachment is required (see 3.11.1.2 and 3.17.2.10).
e. When an auxiliary or third drum is required (see 3.15.7.3).
f. When an overload indicating system is required (see 3.17.1.7).
g. When boom extensions are required, lengths and number required (see 3.17.2.1).
h. When boom jib is required and length required (see 3.17.2.2).
i. When clamshell bucket is required (see 3.17.2.3).
j. When tagline is required (see 3.17.2.4).
k. When dragline bucket is required (see 3.17.2.5).
1. When dragline fairleads are required (see 3.17.2.6).
m. When a wrecking ball is required and size required (see 3.17.2.7).
n. When the pile driver leads and catwalk are required (see 3.17.2.8).
0. When the piledriver hammer is required and size required (see 3.17.2.9).
p. When the backhoe attachment is required (see 3.17.2.11).
q. When paint top coat camouflage green is not required (see 3.20).
r. When the property listed in 3.22 will not be furnished by the Government.
s. When the property listed in 3.23 will not be loaned by the Government.
t. When decontaminating apparatus, portable (DAP), M13 mounting brackets is required (see 3.26).
11. Location of decontaminating apparatus mounting brackets when other than as specified in 3.26.
v* When rifle stowage bracket assembly is required (see 3.27).
w. Locations of rifle stowage bracket assembly when other than as specified in 3.27.
x. Time frame for submitting first article pack (see 5.1).
y. Level of preservation and packing required (see 5.1.1 and 5.2).
z* Any special marking required (see 5.3).
aa. When a transportability report is required (see 3.8 and 6.3).

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6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Description (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements list (DD Form 1423) must be prepared to obtain the data, except where DoD Far Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Paraaraph No.	DID No.	DID title
3.8	DI-PACK-80880	Transportability Report.
3.25.3	DI-H-24029A	Certification, Welding Qualification.
3.25.3	DI-MISC-80875	Welding Procedures.
3.25.3	DI-MISC-80876	Welding Procedure Qualification Test Report.

The above DIDs were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 First article. When a first article inspection is required; the item(s) should be a initial production model. The first article should consist of one or more units. The contracting officer should include specific Instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Lubrications. MIL-STD-838 prescribes the policy for using specification-type products wherever possible and provides specific requirements for potential use of non-standard proprietary products. MIL-STD-838 is implemented by MIL-HDBK-113. The contracting officer should note that unless otherwise authorized by the US ARMY BELVOIR RD&E CTR, ATTN SATBE FL, 10101 GRIDLEY RD, STE 104, FT BELVOIR VA 22060-5818, lubricants, fluids and greases for ground equipment systems must be restricted to those listed under MIL-HDBK-113, chapter 2.

6.5 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.22.

6.6 Government-loaned property. The contracting officer shall arrange to loan the property listed in 3.23.

6.7 Cadmium plated fasteners. The contracting officer should require the contractor to justify the use of cadmium plated fasteners in writing (see 3.4.3).

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Citing standard DoD or commercial practice will not be considered sufficient justification. A specific rationale is required for each specific/different application of cadmium plated fasteners.

6.7.1 Cadmium elated fasteners part numbers. In those instances where cadmium plated fasteners are required and specified (see 3.4.3), drawings and technical manuals should show these cadmium plated fasteners as separate part numbers to preclude the inadvertent use of zinc coated fasteners.

6.8 Subject term (key word) listing.

Lattice Boom Crane
Construction Crane
Crane Backhoe
Crane Shovel
Dragline
Piledriving Rig
Clamshell Bucket
Fairleads
Wrecking Ball
Excavation Crane

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Army - ME

Project 3810-A119

Review activity:
DLA- CS

User activity:
Army - AT .

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-C-52648D(ME)	2. DOCUMENT DATE (YYMMDD) 931203

3. DOCUMENT TITLE Crane, Crawler Mounted, 40 Ton at 12 Foot Radius, 2 Cubic Yard, Diesel-Engine-Driven

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) DSN	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Carolyn B. Johnson	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3468	(2) DSN 654-3468
c. ADDRESS (Include Zip Code) US ARMY BELVOIR RDE CTR SATBE TSE GRIDLEY RD STE 104 FT BELVOIR VA 22060-5818	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: DEFNS QLTY & STDZN OFC 5203 LEESBURG PIKE STE 1403 FLS CHURCH VA 22041-3466 Telephone (703) 756-2340	DSN 289-2340

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